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1 Introduction

Artificial intelligence and robotics have historically followed each other to a certain extent in a complementary manner, but they are two disciplines with largely different goals, focusing respectively on cognitive processes and physical actions. Early artificial intelligence research conceptualized intelligence as a set of abstract, non-physical processes, while robotics emphasized pre-programmed, centrally controlled physical behaviors with limited adaptability. Recent advances have increasingly challenged this distinction and emphasized intelligence as a process that manifests itself in perception, decision-making, and actions in the physical environment.

The study of intelligent collectives inspired by nature serves as a powerful reference for artificial systems. When looking at biological communities such as ant colonies, flocks of birds, or swarms of bees, each individual has myopic vision and limited perception¹. Complex and adaptive behaviors are achieved not through any leader supervising the community, but through the local interactions of individuals with their neighbors. This distributed organization implies that intelligence is, by its nature, decentralized, robust, and scalable. Based on these principles, Radhika Nagpal's work has played an important role in establishing a new paradigm for engineering systems and has demonstrated how simple agents with local rules, capable of self-organization, can exhibit coherent, consistent, and goal-oriented behavior. This perspective lays the foundations for swarm robotics, where intelligence emerges not from a single entity but from the coordinated behavior of numerous autonomous robots with minimal capabilities that act as a single entity.

This bio-algorithmic vision has been the driving force behind Radhika Nagpal's distinguished career, spanning from her doctoral studies at the Massachusetts Institute of Technology to her professorial positions at Harvard and Princeton Universities. Nagpal's groundbreaking contributions, particularly in transforming biological principles into resilient robotic algorithms, have not only redefined the boundaries between computer science and biology but also earned her recognition by Nature as one of the most influential scientists of 2014². Furthermore, fellowships from AAAI and ACM³ underscore

¹Eric Bonabeau et al. *Swarm intelligence: from natural to artificial systems*. Oxford university press, 1999.

²Nature. "Nature's 10: Ten people who mattered this year". In: *Nature* 516 (2014), pp. 311–325.

³Association for Computing Machinery. *2020 ACM Fellows Recognized for Work that Underpins*

Nagpal’s reputation as a figure who bridges the gap between theoretical computer science and experimental robotics and pioneers this field.

2 Theoretical Framework and Synthetic Methodology

Radhika Nagpal takes a bio-algorithmic approach, studying biological systems not as visual models to be imitated, but as information processing systems whose logic can be applied to algorithms. Her research is driven by a fundamental question: What if we could build our own artificial collectives? What if we could build systems where we do not have to build the perfect individual?

2.1 Decentralization and Robustness

Nagpal hypothesizes that natural intelligence is inherently distributed⁴. In systems like termite colonies, collective behavior emerges from the local interactions of “narrow-sighted” individuals, without a central leader⁵. Translating this into engineering, Nagpal focuses on robustness: Unlike centralized robots, where a single failure can bring down the system, Nagpal’s self-organizing swarms function like biological tissues (in the same way that if one or more of the cells involved in muscle repair fail, the muscle tissue continues to be repaired by other cells and the job is completed) - if an individual agent fails, the system immediately adapts and restores its function⁶.

2.2 Synthetic Methodology: Reverse Engineering Nature

Nagpal does not just observe nature; her work is based on synthetic methodology. Nagpal’s approach works like this: She extracts principles from nature, turns these principles into

Today’s Computing Innovations. ACM Website. https://awards.acm.org/award_winners/nagpal_4930808. 2021.

⁴Radhika Nagpal. *Self-organizing Systems Research Group: Research Philosophy*. Harvard University / Princeton University. 2023.

⁵Justin Werfel et al. “Designing collective behavior in a termite-inspired robot construction team”. In: *Science* 343.6172 (2014), pp. 754–758.

⁶Chris R Reid et al. “Army ants dynamically adjust living bridges in response to a cost–benefit trade-off”. In: *Nature communications* 6.1 (2015), pp. 1–7.

algorithms, local rules that can be applied on robots, and observes the behavior of the resulting system and generates new knowledge about which assumptions are sufficient and which are insufficient. So, she doesn't just take it apart; she synthesizes it. In this way, her work creates feedback loops that computer science helps biological discoveries.

2.3 The “Global-to-Local” Programming Paradigm

Nagpal's algorithms are similar to a computer compiler and break the language barrier between humans and robots. A human operator defines an abstract and global goal such as “Create this shape” or “Build this building”, but robots are oblivious to this big picture; they can only control their own actions, see the obstacle in front of them with their sensors or communicate with their neighbor nearby. So, robots don't understand what shape to form; they only recognize local actions like “Go right” or “Stop”.

This is where Nagpal's “Global-to-Local” compilation approach comes in. This system mathematically analyzes the global goal that user gives as input and translates it into simple, low-level rules that each robot can understand and apply.

These algorithms do not give random orders to robots; they mathematically prove and guarantee that those simple rules, applied by thousands of robots looking only at their neighbors, will inevitably lead to the desired global state. The burden of complexity is thus taken away from the hardware and placed entirely on the algorithm, allowing thousands of “mindless” robots to form an “intelligent” whole.

3 Realization of Swarm Intelligence: Case Studies

The theoretical framework and “Global-to-Local” algorithms developed by Nagpal have not only remained on paper but have been validated in the physical world with two landmark projects published in Science. These projects are the most tangible evidence of how simple agents can accomplish complex collective tasks by following local rules. In the following, Kilobots, focusing on “Self-Assembly”, and TERMES, focusing on “Collective Construction”, will be detailed.

3.1 Kilobot: Programmable Swarm and Scalability

In nature, thousands, millions or even trillions of individuals can self-assemble into many different forms just by organizing themselves through local interactions, and researchers in robotics are using collective intelligence in nature as an inspiration, looking at examples such as swarms of ants coming together to traverse difficult terrain.

But until Nagpal's work, there was still a large gap between conceptual design and completed systems. Systems typically had between 10-50 robots⁷⁸, due to the high cost and complex design choices that made production and operations difficult. The lack of large-scale physical systems prevented experimental exploration of self-assembly and other collective intelligence algorithms.

Later, Nagpal and her team presented a system that addressed both the physical and algorithmic challenges of robotic swarms, and the solution came in simplicity. Nagpal's most revolutionary aspect is that she dumbed down the hardware and spread the intelligence across the swarm⁹. In this way, the team created a swarm of a thousand robots that could flexibly reorganize itself to handle highly complex collective behavior, and achieved that with fewer and cheaper components, but with limited capability and reliability.

They used unconventional design choices, such as using vibratory gliding instead of traditional wheels to move and reflecting infrared light off the ground for functions such as communication and measuring the distance between robots. The self-assembly algorithm developed consists of 3 primitive collective behaviors:

- Edge-following: Robots move around the swarm over the edge, measuring the distance between them and other robots on the outer edges.

- Gradient Formation¹⁰: With a signal emitted from a seed robot, each robot calculates its distance from the center.

- Localization: Robots determine their local coordinates based on their distance from their neighbors.

Although the robots are able to perform the given tasks and form the desired shapes

⁷Francesco Mondada et al. "The swarm-bot system". In: *Autonomous Robots* 17 (2004), pp. 193–221.

⁸Gilles Caprari et al. "Alice: A compact autonomous mobile robot for swarm robotics research". In: *Haptics-e*. 2002.

⁹Michael Rubenstein et al. "Programmable self-assembly in a thousand-robot swarm". In: *Science* 345.6198 (2014), pp. 795–799.

¹⁰Ibid.

thanks to these basic principles and constant communication with their neighbors, unexpected failures are inevitable in an environment with 1000 robots. To maintain robustness in this system, Nagpal has developed a key technique called Cooperative Monitoring. If a robot breaks down (for example, if its motor fails), it gives a signal to its neighbors, and the other robots get around it and continue the task. This mechanism is the artificial equivalent of the resilience of biological systems and guarantees that the pattern can be completed.

3.2 TERMES: Stigmergy and Autonomous Construction

Complex systems are characterized by many independent components that produce high-level outcomes from low-level rules. However, the main challenge here is the “Inverse Problem”: Determining which simple rules must be provided to achieve these desired high-level outputs still remains as a poorly understood issue.

In their work published in *Science* magazine, Nagpal and her colleagues, inspired by termites in nature, presented multi-agent construction systems that solve this inverse problem. The user explicitly specifies the structure to be built, and the system automatically generates low-level local rules for independent climbing robots that guarantee the construction of the structure. This work aims at engineering complex systems that achieve specific human-designed goals.

Unlike human-made projects, which require careful planning and strict supervision, in animals (especially in termites which are the most striking example), the same work is carried out by millions of individuals acting independently and in a decentralized way. Nagpal’s system also uses this biological principle: Stigmergy. There is no direct communication between robots; communication is achieved through a shared environment¹¹(e.g., the structure of the mound itself or pheromone trails left). In other words, robots perceive the current state of the structure and make their next move accordingly (e.g. “If there is a gap here, put a brick”).

The robots used in the system are equipped with limited capabilities. They can move back and forth, turn around and, most importantly, climb up to the height of a brick, so that they can build stairs and towers on top of the structure they are building. However, their perception is extremely limited (myopic sensing); robots can only see the bricks and

¹¹Werfel et al., op. cit.

other robots directly in front of them. They do not know the current state of the whole building and what other robots are doing in the distance. This is where “Stigmergy” comes in, and the magic of this system is that it doesn’t need to know the “big picture”.

In order to create a smooth structure despite this limitation of local perception, the building process takes place on a “non-overlapping grid pattern”¹². The movement areas of the bricks and robots are divided into predetermined squares, which keeps error to a minimum. As a result, Nagpal proved that without human intervention or a centralized leader, an autonomous swarm following only local rules can build complex 3D structures much larger than themselves.

4 Impact on Academic Culture and Leadership

Radhika Nagpal’s place in the world of science is not limited to being a great engineer. She is also a leader against the toxic culture of competition, burnout and gender inequality in academy. She is a living example, especially for women working in science and engineering, that success is not only about hard work, but also about living a smart and balanced life despite marriage and family life.

Nagpal expressed her philosophy in her manifesto “The Awesomest 7-Year Postdoc”¹³, which went viral in the academic world.

4.1 The “7-Year Postdoc” Philosophy: Conquering Fear

In the most prestigious (R1) universities in the world, such as Harvard, there is a 7-year probationary period for young professors called “Tenure-Track”. This is usually a period of intense pressure, where people work 80-hour days to get tenure, burn out, and are in constant fear of being fired.

Nagpal sort of rebelled against this fearful system. When she started at Harvard, she put a note on her desk saying, “This is a 7-year Postdoc”. Her philosophy was: “I didn’t come here to stay forever. I came here to hang out for 7 years with the smartest and coolest folks in the world and play games in a great lab. If I don’t get tenure at the end of 7 years, at least I’ve had a great 7 years.”

¹²Ibid.

¹³Radhika Nagpal. *The Awesomest 7-Year Postdoc or: How I Learned to Stop Worrying and Love the Tenure-Track Faculty Life*. Scientific American Blog Network. 2013.

This perspective helped Nagpal relieve her from the pressure to succeed. Nagpal refused to sit on tedious committees just to get tenure, ignored the “to-do list for tenure,” and focused only on the projects she truly wanted to work on. Ironically, the moment she stopped chasing success, success chased her, and she received tenure at Harvard.

4.2 “Being the Best Whole Person”

Nagpal has also shattered the “Superhero” myth in the academic world. In response to Larry Summers, who also served as Secretary of the Treasury, implying while he was Harvard President that “Women can’t reach the top because they don’t want to work 80 hours a week”¹⁴, Nagpal did her own math. As a mother of two, she realized that working 80 hours a week was impossible without abandoning her children. Wanting to be the most supportive spouse, the most dedicated mother, and the best scientist she could be, Nagpal set her own rules¹⁵:

- Fixed Working Hours: Not working more than 50 hours a week.
- Weekend Ban: Not checking emails from Friday evening to Monday morning.
- “No” Quota: Accept only a limited number of trips and committee assignments per year and say “No” to everything once the quota is full.

Nagpal calls this “Being the Best Whole Person.” In her view, being a good scientist should not require giving up being a good mother or a healthy individual.

4.3 Leadership and Mentoring

Nagpal’s attitude has made her not only a technical authority but also a cultural icon. She mentors young researchers by saying “Don’t let the system crush you, make your own rules”. This attitude earned her the Anita Borg Early Career Award in 2010¹⁶ and the McDonald Mentoring Award in 2015¹⁷. Nagpal is also now part of academic

¹⁴Daniel J. Hemel. “Summers’ Remarks on Women Draw Fire”. In: *The Boston Globe* (2005). Available at: <https://www.thecrimson.com/article/2005/1/14/summers-comments-on-women-and-science/>.

¹⁵Nagpal, op. cit.

¹⁶AnitaB.org. *Radhika Nagpal: 2010 Anita Borg Early Career Award*. AnitaB.org Website. <https://ssr.princeton.edu/people/radhika-nagpal>. 2010.

¹⁷Harvard SEAS. *Nagpal honored for excellence in mentoring and advising*. Harvard John A. Paulson School of Engineering and Applied Sciences News. Available at: <https://seas.harvard.edu/news/>

history by arguing that science is not just about doing research in the lab, but also about protecting people’s mental health. Nagpal’s biggest projects (Kilobots and Termes) were born precisely during that 7-year period when her academic career was at its riskiest. This is the proof that her strategy of rejecting fear and focusing on curiosity and passion was the right one.

5 Summary

Radhika Nagpal’s career and work represent not only technological progress in robotics and computer science, but also a change in mindset. She has applied artificial intelligence not to achieve a single supercomputer, but to obtain collective intelligence—one of nature’s most powerful strategies—by breaking it down into thousands of simple parts that behave stupidly on their own.

As detailed in this paper, one of Nagpal’s greatest legacies is his “Global-to-Local” programming paradigm. This made it possible for human operators to manage thousands of robots as a single body, without the need to tell each robot exactly what to do. The self-assembling swarm robots of the Kilobots project and the leaderless construction robots of the TERMES project are the most concrete proofs that theoretical computer science can leave the laboratory and work in the physical world.

However, Nagpal’s influence goes beyond the laboratory. She is a scientist inspired by nature to solve biological mysteries and a role model who has managed to remain human in the competitive environment of the relentless academic world. The success she achieved by rejecting pressure and playing by her own rules serves as a roadmap for future scientists, just as valuable as her technical papers.

Looking to the future, it is exciting to see where the foundations laid by Nagpal will extend. The next generations of those simple robots forming the letter “K” in the lab today, will tomorrow be swarms of nano-robots searching for cancer cells in our veins, autonomous fleets cleaning up pollution in the oceans, robotic workers going to extraterrestrial locations like Mars and building bases there before humans arrive. What Nagpal teaches us is this: Intelligence does not have to be gathered in a single center; intelligence is wherever there is cooperation.

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7 Appendix

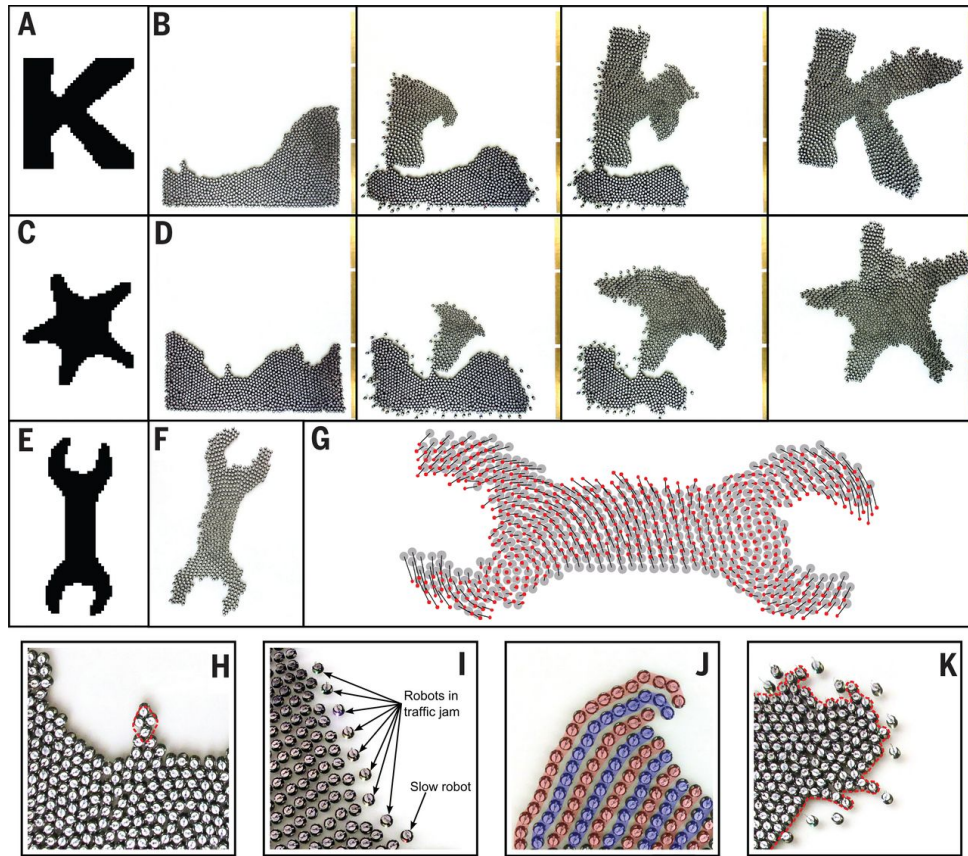


Figure G.1: Experimental demonstration of self-assembly with a thousand-robot swarm. The Kilobots autonomously organize into complex 2D shapes, such as the letter 'K' and a wrench, using local interactions and gradient formation.